Chap 4. Integration against rough paths

Choi, Min-Jun

Seoul National University

minjunchoi@snu.ac.kr

April 2, 2020

Overview

1. Introduction
   - Young Integral
2. Integration of 1-form
   - Sewing Lemma
3. Integration of controlled rough paths
   - The general rough path integral

Introduction

The aim of this chapter is to give a meaning to the expression $\int Y_t dX_t$ a suitable class of integrands Y, integrated against a rough path X.

[You36] The Young Integral

Let V be a Banach space and $\alpha+\beta>1$. If $X\in\mathcal{C}^{\alpha}\left([0,T],V\right)$ and $Y\in\mathcal{C}^{\beta}\left([0,T],V\right)$, then the Young integral exists, that is,

$$\int_0^1 Y_t dX_t = \lim_{|\mathcal{P}|\to 0}\sum_{[s,t]\in\mathcal{P}} Y_s X_{s,t},$$

where $\mathcal{P}$ denotes a partition of $[0,1]$ and $|\mathcal{P}|$ denotes the length of the largest element of $\mathcal{P}$. Moreover Young integral has the bound

$$\left|\int_s^t Y_r dX_r - Y_s X_{s,t}\right| \leq C||Y||_{\beta}||X||_{\alpha}\left|t-s\right|^{\alpha+\beta}. \tag{1}$$

Here, our main interest is to reduce $\alpha>\frac{1}{2}$ in the case $\alpha=\beta$.
The easiest way for a function Y to "looks like X" is to have $Y_t = F\left(X_t\right)$ for some sufficiently smooth $F:V\to\mathcal{L}\left(V,W\right)$, called a 1-form.

Integration of 1-form

Let $Y = F(X)$ and $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^{\alpha}([0, T], V)$. When $F : V \to \mathcal{L}(V, W)$ is in C_b^2, a Taylor approximation gives

$$F(X_r) \approx F(X_s) + DF(X_s)\, X_{s,r}$$

for r in some (small) interval $[s, t]$. Then since the definition of Young integral, $F(X_r) \approx F(X_s)$ for $r \in [s, t]$ and then

$$\int_0^1 F(X_s) d\mathbf{X}_s \approx \sum_{[s,t] \in \mathcal{P}} \left[F(X_s)\, X_{s,t} + DF(X_s)\, \mathbb{X}_{s,t} \right],$$

which is a good enough approximation.

Why should this be good enough?
The intuition is as follows: given $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, neither $|X_{s,t}| \sim |t-s|^{\alpha}$ nor $|\mathbb{X}_{s,t}| \sim |t-s|^{2\alpha}$ in the above sum will be negligible as $|\mathcal{P}| \to 0$.
Moreover, $\mathbb{X}^{(3)}_{s,t} \sim |t-s|^{3\alpha} = o(|t-s|)$. So, we will see that this limit,

$$\int_0^1 F(X_s) d\mathbf{X}_s = \lim_{|\mathcal{P}| \to 0} \sum_{[s,t] \in \mathcal{P}} \left[F(X_s) X_{s,t} + DF(X_s) \mathbb{X}_{s,t} \right], \qquad (2)$$

does exists and call it *rough integral*.

Integration of 1-form

Lemma 4.1

Let $F : V \to \mathcal{L}(V, W)$ be a C_b^2 function and let $(X, \mathbb{X}) \in \mathscr{C}^\alpha$ for some $\alpha > \frac{1}{3}$. Set $Y_s := F(X_s)$, $Y'_s := DF(X_s)$ and $R^Y_{s,t} := Y_{s,t} - Y'_s X_{s,t}$. Then

$$Y, Y' \in \mathcal{C}^\alpha \text{ and } R^Y \in \mathcal{C}^{2\alpha}.$$

More precisely, we have the estimates

$$\begin{aligned} ||Y||_\alpha &\le ||DF||_\infty ||X||_\alpha, \\ ||Y'||_\alpha &\le ||D^2F||_\infty ||X||_\alpha, \\ ||R^Y||_{2\alpha} &\le \frac{1}{2} ||D^2F||_\infty ||X||^2_\alpha. \end{aligned}$$

Proof. $\mathcal{C}_b^2$ regularity of F implies that F and DF are both Lipschitz continuous with Lipschitz constants $||DF||_\infty$ and $||D^2F||_\infty$ respectively. The α-Hölder bounds on Y and Y' are then immediate. For the remainder term, consider the function

$$[0,1] \ni \xi \mapsto F(X_s + \xi X_{s,t}).$$

A Taylor expansion, with intermediate value remainder, yields $\xi \in (0,1)$ such that

$$R^Y_{s,t} = F(X_t) - F(X_s) - DF(X_s)X_{s,t} = \frac{1}{2}D^2F(X_s + \xi X_{s,t})(X_{s,t}, X_{s,t}).$$

The claimed 2α-Hölder estimae, in the sense that $|R^Y_{s,t}| \lesssim |t-s|^{2\alpha}$, then follows at once.

$\square$

Integration of 1-form

Let $Z_t := \int_0^t Y_r dX_r$. Then by Young's inequaility (1), one has

$$Z_{s,t} = Y_s X_{s,t} + o(|t-s|).$$

That is, the function $\Xi_{s,t} := Y_s X_{s,t}$ fully determines $Z_{s,t}$. Therefore we want to find Ξ, such that $Z = \mathcal{I}\Xi$ is a well defined image of Ξ under an abstract integration map $\mathcal{I}$.

Integration of 1-form

Definition

We define the space $\mathcal{C}_2^{\alpha,\beta}([0,T],W)$ of functions Ξ from the simplex $0 \leq s \leq t \leq T$ into W such that $\Xi_{t,t} = 0$ and

$$||\Xi||_{\alpha,\beta} := ||\Xi||_\alpha + ||\delta\Xi||_\beta < \infty,$$

where $||\Xi||_\alpha = \sup_{s<t} \frac{|\Xi_{s,t}|}{|t-s|^\alpha}$ as usual and also

$$\delta\Xi_{s,u,t} = \Xi_{s,t} - \Xi_{s,u} - \Xi_{u,t}, \;\; ||\delta\Xi||_\beta = \sup_{s<u<t} \frac{|\delta\Xi_{s,u,t}|}{|t-s|^\beta}.$$

Lemma 4.2 (Sewing lemma)

Let α and β be such that $0 < \alpha \leq 1 < \beta$. Then, there exists a (unique) continuous map $\mathcal{I} : \mathcal{C}_2^{\alpha,\beta}([0,T],W) \to \mathcal{C}^{\alpha}([0,T],W)$ such that $(\mathcal{I}\Xi)_0 = 0$ and

$$|(\mathcal{I}\Xi)_{s,t} - \Xi_{s,t}| \leq C|t-s|^{\beta}, \tag{3}$$

where C only depends on β and $||\delta\Xi||_{\beta}$.

Integration of 1-form

Proof) For a partition $\mathcal{P}$ of $[s, t]$, set

$$\int_{\mathcal{P}} \Xi := \sum_{u,v \in \mathcal{P}} \Xi_{u,v}$$

We want to show that

$$\lim_{|\mathcal{P}| \to 0} \int_{\mathcal{P}} \Xi$$

exists; this will define $\mathcal{I}\Xi$.

Step 1. We show that given any partition $\mathcal{P}$, we have

$$\sup_{\mathcal{P} \subset [s,t]} \left| \Xi_{s,t} - \int_{\mathcal{P}} \Xi \right| \leq C ||\delta \Xi||_{\beta} |t - s|^{\beta}. \tag{4}$$

Consider a partition $\mathcal{P}$ of $[s, t]$ and let $r \geq 1$ be the number of intervals in $\mathcal{P}$. When $r \geq 2$ there exists $u \in [s, t]$ such that $[u_-, u]$, $[u, u_+] \in \mathcal{P}$ for some u_-, u_+ and

$$|u_+ - u_-| \leq \frac{2}{r-1} |t - s|.$$

Integration of 1-form

Removing the point u from $\mathcal{P}$, we obtain

$$\begin{aligned}\left|\int_{\mathcal{P}} \Xi - \int_{\mathcal{P}\setminus\{u\}} \Xi\right| &= \left|\sum_{u,v\in\mathcal{P}} \Xi_{u,v} - \sum_{u,v\in\mathcal{P}\setminus\{u\}} \Xi_{u,v}\right| \\ &= \left|\Xi_{u_-,u} + \Xi_{u,u_+} - \Xi_{u_-,u_+}\right| \\ &= \left|\delta\Xi_{u_-,u,u_+}\right| \\ &\leq ||\delta F||_\beta \left(\frac{2}{r-1}|t-s|\right)^\beta\end{aligned}$$

Integration of 1-form

Further, we can see that if there are more than 3 elements in $\mathcal{P}$, i.e., $r \geq 3$, there exists two points $u, v' \in \mathcal{P}$ such that

$$\begin{aligned}
&\left|\Xi_{s,t} - \int_{\mathcal{P}} \Xi\right| \\
&\leq \left|\Xi - \int_{\mathcal{P}\setminus\{u\}} \Xi\right| + \left|\int_{\mathcal{P}\setminus\{u\}} \Xi - \int_{\mathcal{P}} \Xi\right| \\
&\leq \left|\Xi - \int_{\mathcal{P}\setminus\{u,v'\}} \Xi\right| + \left|\int_{\mathcal{P}\setminus\{u,v'\}} \Xi - \int_{\mathcal{P}\setminus\{u\}} \Xi\right| + \left|\int_{\mathcal{P}\setminus\{u\}} \Xi - \int_{\mathcal{P}} \Xi\right|
\end{aligned}$$

By iterating this procedure, selecting a new point u to remove each time, we get that the difference between $\Xi_{s,t}$ and $\int_{\mathcal{P}} \Xi$ biggest, we see that,

Integration of 1-form

$$\sup_{\mathcal{P}\subset[s,t]}\left|\Xi_{s,t}-\int_{\mathcal{P}}\Xi\right| \leq ||\delta F||_{\beta}\sum_{i=2}^{r}\left(\frac{2}{i-1}|t-s|\right)^{\beta}$$
$$\leq 2^{\beta}\zeta(\beta)||\delta\Xi||_{\beta}|t-s|^{\beta}$$

where $\zeta(\beta)=\sum_{r=1}^{\infty}\frac{1}{r^{\beta}}$ is the Riemann zeta function. It then remains to show that

$$\sup_{|\mathcal{P}|\vee|\mathcal{P}'|<\epsilon}\left|\int_{\mathcal{P}}\Xi-\int_{\mathcal{P}'}\Xi\right| \to 0 \ \text{ as } \ \epsilon\to 0.$$

Integration of 1-form

Step 2. Take any two partitions $\mathcal{P}$ and $\mathcal{P}'$; we can suppose without loss of generality that $\mathcal{P}'$ refines $\mathcal{P}$. Then since every $[s,t] \in \mathcal{P}$ either belongs to $\mathcal{P}'$ or is divided in sub-intervals that belong to $\mathcal{P}'$, we have

$$\begin{aligned}\left|\int_{\mathcal{P}} \Xi - \int_{\mathcal{P}'} \Xi\right| =& \left|\sum_{[u,v]\in\mathcal{P}} \left(\Xi_{u,v} - \int_{\mathcal{P}'\cap[u,v]} \Xi\right)\right| \\ \leq& 2^{\beta}\zeta(\beta)||\delta\Xi||_{\beta} \sum_{[u,v]\in\mathcal{P}} |u-v|^{\beta}\end{aligned}$$

which proves that

$$\sup_{|\mathcal{P}|\vee|\mathcal{P}'|<\epsilon} \left|\int_{\mathcal{P}} \Xi - \int_{\mathcal{P}'} \Xi\right| \to 0 \quad \text{as} \quad \epsilon \to 0.$$

Therefore the limit

$$\lim_{|\mathcal{P}|\to 0} \int_{\mathcal{P}} \Xi =: (\mathcal{I}\Xi)_{s,t}$$

exists and defines a linear function $\mathcal{I}$ that, thanks to the estimate above is bounded, hence continuous.

The property $(\mathcal{I}\Xi)_0 = 0$ is trivial. Moreover, $\delta(\mathcal{I}\Xi)_{s,u,t} = 0$ for any $s, u, t \in [0, T]$. Estimate (3) follows immediately from (4).

For uniqueness, if $\mathcal{I}$ and $\tilde{\mathcal{I}}$ are two continuous linear map satisfying (3), there difference $\mathcal{I} - \tilde{\mathcal{I}}$ satisfies $\left(\mathcal{I} - \tilde{\mathcal{I}}\right)_0 = 0$ and $\left(\mathcal{I} - \tilde{\mathcal{I}}\right)_{s,t} \precsim |t-s|^{\beta}$. Since $\beta > 1$ by assumption, it follows immediately that $\mathcal{I} - \tilde{\mathcal{I}}$ vanishes identically.

$\square$

Integration of 1-form

Theorem 4.4 (Lyons).

Let $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha([0, T], V)$ for some $T > 0$ and $\alpha > \frac{1}{3}$, and $F : V \to \mathcal{L}(V, W)$ be a C_b^2 function. Then, the rough integral defined

$$\int_0^1 Y_t dX_t = \lim_{|\mathcal{P}| \to 0} \sum_{[s,t] \in \mathcal{P}} (F(X_s) X_{s,t} + DF(X_s) \mathbb{X}_{s,t})$$

exists and one has the bound

$$\left| \int_s^t F(X_r)\, d\mathbf{X}_r - F(X_s) X_{s,t} - DF(X_s) \mathbb{X}_{s,t} \right| \lesssim ||F||_{C_b^2} \left(||X||_\alpha^3 + ||X||_\alpha ||\mathbb{X}||_{2\alpha} \right) |t - s|^{3\alpha},$$

where the proportionality constant depends only on α.

Integration of 1-form

Furthermore, the indefinite rough integral is α-Hölder continuous on $[0, T]$ and we have the following quantitative estimate,

$$\| \int_0^{\cdot} F(X)\, d\mathbf{X} \|_\alpha \leq C \|F\|_{C_b^2} \left(|||\mathbf{X}|||_\alpha \vee |||\mathbf{X}|||_\alpha^{1/\alpha} \right),$$

where the constant C only depends on T and α and can be chosen uniformly in $T \leq 1$. Furthermore, $|||\mathbf{X}|||_\alpha = ||X||_\alpha + \sqrt{||\mathbb{X}_{2\alpha}||}$ denotes again the homogeneous α-Hölder rough path norm.

Definition 4.6

Given a path $X \in \mathcal{C}^\alpha([0,T],V)$, we say that $Y \in \mathcal{C}^\alpha([0,T],\overline{W})$ is *controlled* by X if there exists $Y' \in \mathcal{C}^\alpha([0,T],\mathcal{L}(V,\overline{W}))$ so that the remainder term R^Y given implicitly through the relaion

$$Y_{s,t} = Y'_s X_{s,t} + R^Y_{s,t},$$

satisfies $||R^Y||_{2\alpha} < \infty$. This defines the space of *controlled rough paths*,

$$(Y, Y') \in \mathscr{D}^{2\alpha}_X([0,T],\overline{W}).$$

Although Y' is not, in general, uniquely determined from Y we call any such Y' the *Gubinelli derivative* of Y (with respect to X).

Integration of controlled rough paths

Theorem 4.10 (Gubinelli)

Let $T > 0$, $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha([0,T], V)$ for some $\alpha > \frac{1}{3}$, and $(Y, Y') \in \mathscr{D}_X^{2\alpha}([0,T], \mathcal{L}(V, W))$. Then there exists a constant C depending only on T and α (and C can be chosen uniformly over $T \in (0,1]$) such that
(a) The integral defined by

$$\int_0^1 Y d\mathbf{X} = \lim_{|\mathcal{P}| \to 0} \sum_{[s,t] \in \mathcal{P}} \left(Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t} \right)$$

exists and, for every pair s, t, one has the bound

$$\begin{aligned} &\left| \int_s^t Y_r d\mathbf{X}_r - Y_s X_{s,t} - Y'_s \mathbb{X}_{s,t} \right| \\ \leq & C \left(||X||_\alpha ||R^Y||_{2\alpha} + ||\mathbb{X}||_{2\alpha} ||Y'||_\alpha \right) |t-s|^{3\alpha} . \end{aligned}$$

Integration of controlled rough paths

(b) The map from $\mathscr{D}_X^{2\alpha}([0,T],\mathcal{L}(V,W))$ to $\mathscr{D}_X^{2\alpha}([0,T],W)$ given by

$$(Y,Y') \mapsto (Z,Z') := \left(\int_0^{\cdot} Y_t d\mathbf{X}_t, Y\right),$$

is a continuous linear map between Banach spaces and one has the bound

$$||Z,Z'||_{X,2\alpha} \leq ||Y||_\alpha + ||Y'||_{L^\infty}||\mathbb{X}||_{2\alpha} + C\left(||X||_\alpha||R^Y||_{2\alpha} + ||\mathbb{X}||_{2\alpha}||Y'||_\alpha\right).$$

*$||Y,Y'||_{X,2\alpha} := ||Y'||_\alpha + ||R^Y||_{2\alpha}$

Integration of controlled rough paths

Proof) We want to apply the sewing lemma with $\Xi_{s,t} = Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t}$, therefore we need to check that $||\delta F||_\beta < \infty$ for some $\beta > 1$. By Chen's relation,

$$
\begin{aligned}
\delta F_{s,u,t} =& F_{s,t} - F_{s,u} - F_{u,t} \\
=& Y_s X_{s,t} - Y_s X_{s,u} - Y_u X_{u,t} + Y'_s \mathbb{X}_{s,t} - Y'_s \mathbb{X}_{s,u} - Y'_u \mathbb{X}_{u,t} \\
=& - Y_{s,u} X_{u,t} + Y'_s \left(\mathbb{X}_{s,t} - \mathbb{X}_{s,u} \right) - Y'_u \mathbb{X}_{u,t} \\
=& - \left(Y'_s X_{s,u} + R^Y_{s,u} \right) X_{u,t} + Y'_s (\mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t}) - Y'_u \mathbb{X}_{u,t} \\
=& Y'_{u,s} \mathbb{X}_{u,t} - R^Y_{s,u} X_{u,t}.
\end{aligned}
$$

Integration of controlled rough paths

Therefore

$$\begin{aligned} |\delta F_{s,u,t}| =& ||Y'_{u,s}||_\alpha |s-u|^\alpha ||\mathbb{X}_{u,t}||_{2\alpha}|t-u|^{2\alpha} \\ &+ ||R^Y_{s,u}||_{2\alpha}|u-s|^{2\alpha}||X_{u,t}||_\alpha |t-u|^\alpha, \end{aligned}$$

hence,

$$||\delta F||_{3\alpha} \le ||Y'||_\alpha ||||X||_{2\alpha} + ||R^Y||_{2\alpha}||X||_\alpha =: C_{X,Y}$$

and we can apply the sewing lemma exactly if $\alpha > \frac{1}{3}$. The first estimate then follows immediately from step 1 in the proof of the sewing lemma. For (b), the estimate in (a) implies that

$$|Z_t - Z_s| = \left| \int_s^t Y_r dX_r \right| \le |Y_s X_{s,t}| + \left| Y'_s \mathbb{X}_{s,t} \right| + C_{X,Y}|t-s|^{3\alpha}$$

where $C_{X,Y}$ is the bound for $||\delta F||$ above.

This can be bounded by

$$||Y||_\infty||X||_\alpha|t-s|^\alpha + ||Y'||_\infty||\mathbb{X}||_{2\alpha}|t-s|^{2\alpha} + C_{X,Y}|t-s|^{3\alpha}$$

yielding

$$||Z||_\alpha \lesssim ||X||_\alpha\left(||Y||_\alpha + ||R^Y||_{2\alpha}\right) + ||\mathbb{X}||_{2\alpha}||Y'||_\infty,$$

thus $Z \in \mathcal{C}^\alpha$.
By (a), $Z_{s,t} = \int_s^t Y_r dX_r = Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t} + \tilde{R}_{s,t}$ and $||\tilde{R}||_{3\alpha} < \infty$; moreover, $|Y'_s\mathbb{X}_{s,t}| \leq ||Y'||_\infty||\mathbb{X}||_{2\alpha}|t-s|^{2\alpha}$ thus $Z \in \mathscr{D}_X^{2\alpha}$ and $Z' = Y$ (the term $Y'_s\mathbb{X}_{s,t}$ is the 2α-remainder.).
The last estimate for $||Z,Z'||_{X,2\alpha} = ||Y||_\alpha + ||R^Z||_{2\alpha}$ follows immediately.
Finally, continuity is given by the sewing lemma.

$\square$

Thank you for your attention!